

Module -III

Biochemistry



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Module -II

Biochemistry

- Cell Metabolism
 - Aerobic and anaerobic respiration
 - Glycolysis
 - Krebs cycle
- Electron transport chain
- Enzymes and Catalysis
- Cell Communication and Signalling

Module -II**What is Energy ?**

- Energy can be defined as the ability to do work.
- If an object or organism does work (exerts a force over a distance to move an object) the object or organism uses energy.
- In addition to using energy to do work, objects gain energy because work is being done on them.
- Forms of energy –
 - Thermal energy
 - Mechanical energy
 - Chemical energy
 - Electrical energy
 - Bioenergy

Characters of living things

- | | |
|-------------------------------|-----------------------|
| • Survivability | • Metabolism |
| • Adaptations | • Movement |
| • Responsiveness | • Reproduction |
| • Genetic composition | • Development |
| • Mutation | • Growth |
| • Evolution | • Respiration |
| • Genetic variations | |
| • Cellular composition | |

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Bioenergetics

- Bioenergetics is a field of Biochemistry and Cell Biology.
- The study of the transformation of energy in living organisms.

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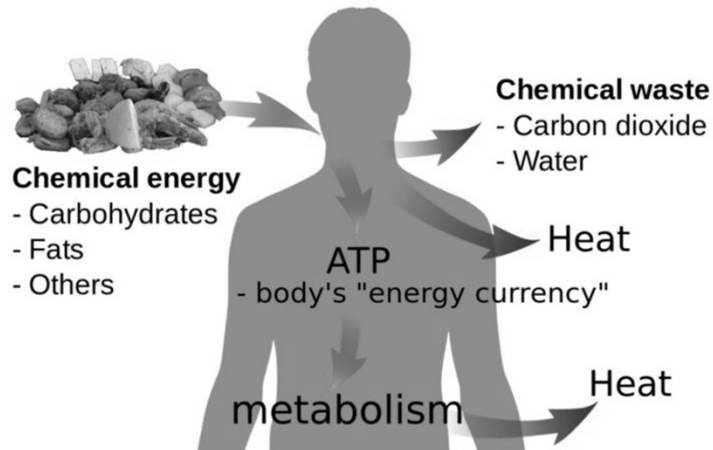
Bioenergetics



- Life is an energy intensive process.
- It takes energy to operate muscles, extract wastes, make new cells, heal wounds, even to think.

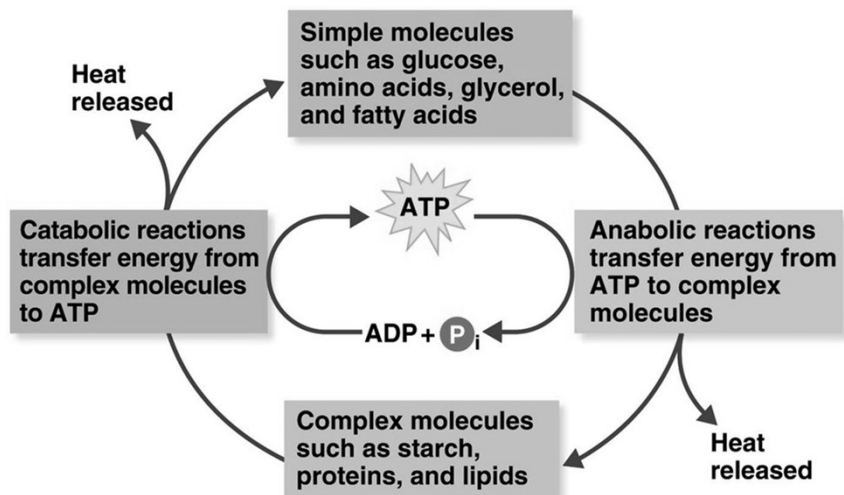
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Energy and human life



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Energy and Human Life



Module -II**Gibbs Free Energy**

- In thermodynamics, the Gibbs free energy is a thermodynamic potential that can be used to calculate the maximum of reversible work that may be performed by a thermodynamic system at a constant temperature and pressure.
- Gibbs free energy is a measure of chemical energy.
- All chemical systems tend naturally toward states of minimum Gibbs free energy

$$G = H - TS$$

Where:

G = Gibbs Free Energy

H = Enthalpy (heat content)

T = Temperature in Kelvins

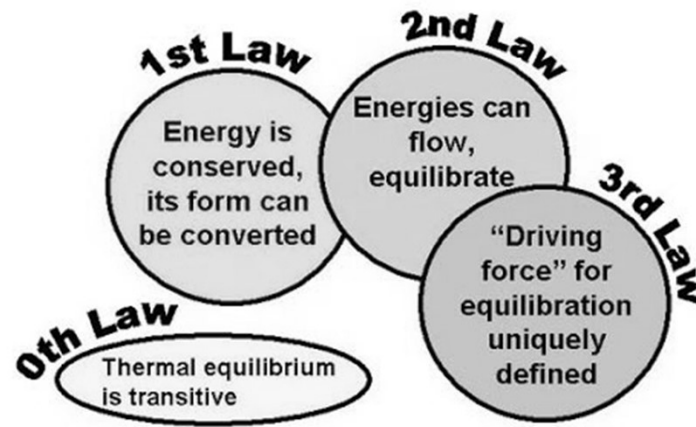
S = Entropy (can think of as randomness)

Module -II**What is thermodynamics?**

- Thermodynamics is the study of energy transformations that occur in a collection of matter.
- Thermodynamics is concerned with the storage, transformation and dissipation of energy.
- Cells store energy, they transform it and they dissipate to drive unfavorable reactions.

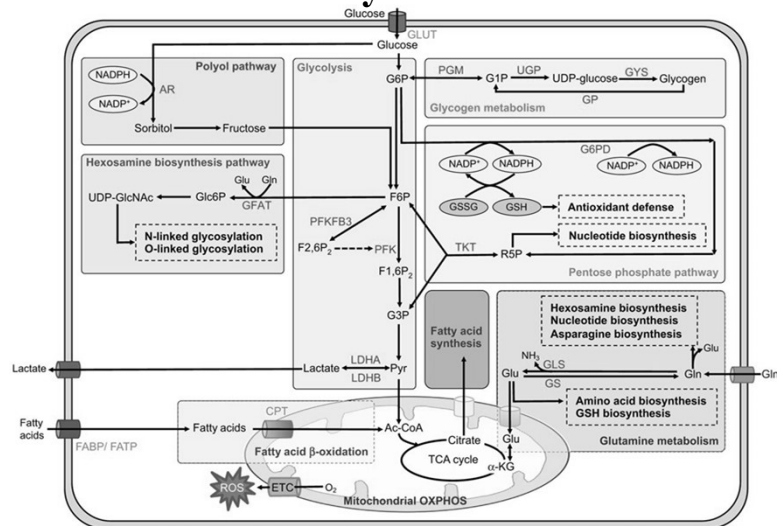
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The Thermodynamic Laws



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Biochemical Pathways



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Cell Metabolism

- Cellular metabolism comprises the controlled biochemical processes that occur within a cell to maintain life. It is generally divided in, anabolic processes (the build-up of macromolecules) and catabolic processes (the breakdown of macromolecules). To maintain constant cellular conditions in response to changes in the environment (homeostasis) a cell regulates its specific needs by the fine-tuning of these metabolic pathways..
- These are enzyme-linked / enzyme-catalyzed reactions.

Poster-Metabolic Pathways

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Metabolism

- Metabolism is the set of life-sustaining chemical transformations or reactions taking place within the cells of living organisms.
- These are enzyme-linked / enzyme-catalyzed reactions.

Purpose

- Conversion of food to energy to run cellular processes.
- Conversion of food to form building blocks e.g. Proteins, Lipids, Carbohydrates, Nucleic Acids.
- To eliminate waste material generated in side the cell e.g. nitrogenous wastes.

Poster-Metabolic Pathways

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Metabolism

- **Catabolism** – Those processes whose major function is the generation of chemical energy in forms suitable for the mechanical and chemical processes of the cell.

- **Anabolism** – Those processes which utilize the energy generated by catabolism for the biosynthesis of organic molecules.

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Metabolism*Breaking down*

- **Catabolism** – Those processes whose major function is the generation of chemical energy in forms suitable for the mechanical and chemical processes of the cell.

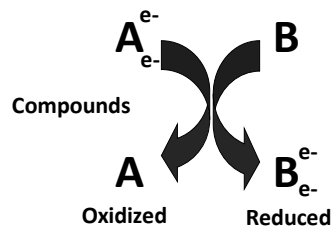
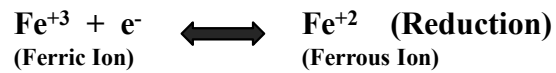
Building up

- **Anabolism** – Those processes which utilize the energy generated by catabolism for the biosynthesis of organic molecules.

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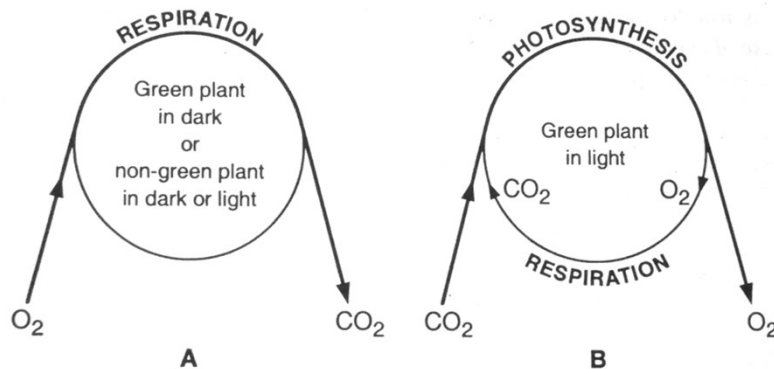
Biological Oxidation – Reduction Reactions

- Several types of chemical reactions are involved in energy production.
- **Oxidation** – Loss of e^- , dehydrogenation
- **Reduction** – Gain of e^- , hydrogenation



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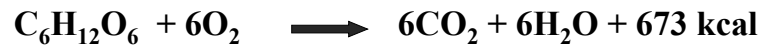
Biological Oxidation – Reduction Reactions



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Respiration

- The process of gaseous exchange where O_2 is usually absorbed from the atmosphere and CO_2 is released. The organic matter is broken down (catabolism) and the energy is released.



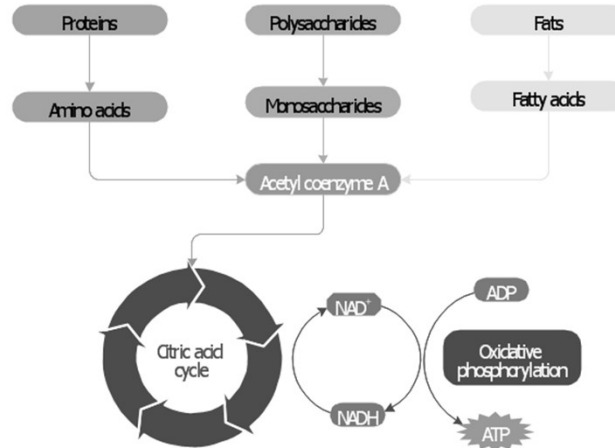
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Respiration

Respiration	Combustion or Burning
Biological Process	Chemical Process
Normal temperature	Higher temperature
Step wise breakdown	Fast process
Enzymes involved	Enzymes not involved
Generate Energy – ATP	Generate Heat & Light

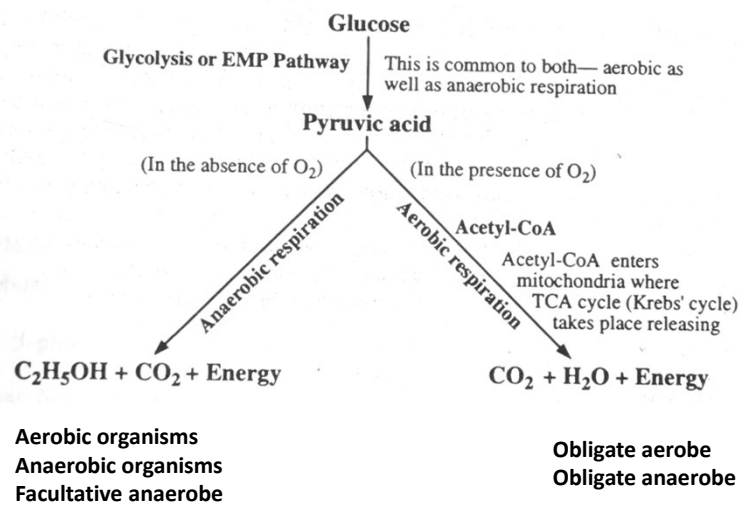
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Catabolism schematic

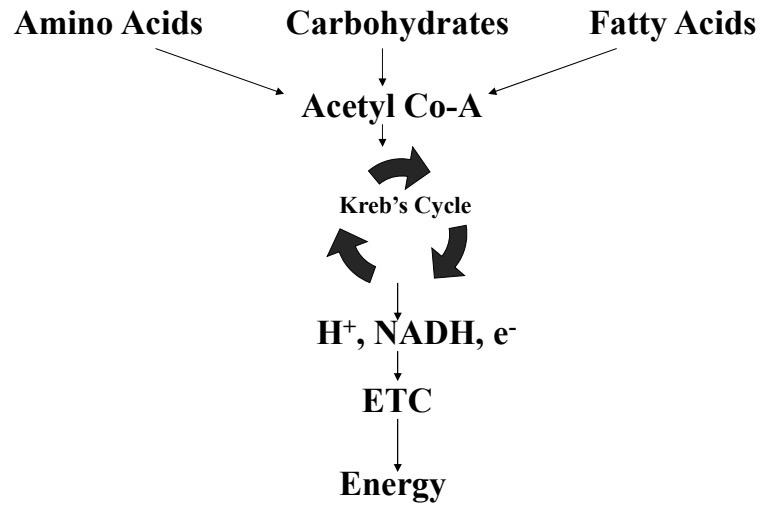


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Mechanism of Respiration



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Mechanism of Respiration

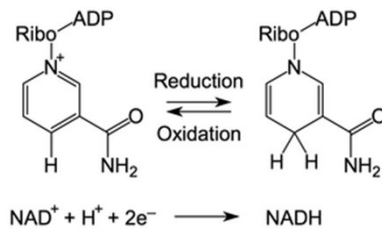
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Universal electron carrier

NAD – Nicotinamide adenine dinucleotide

NADP - Nicotinamide adenine dinucleotide phosphate

FAD – Flavin adenine dinucleotide



NAD.2H / NADH₂ / NADH

FAD.2H / FADH₂ / FADH

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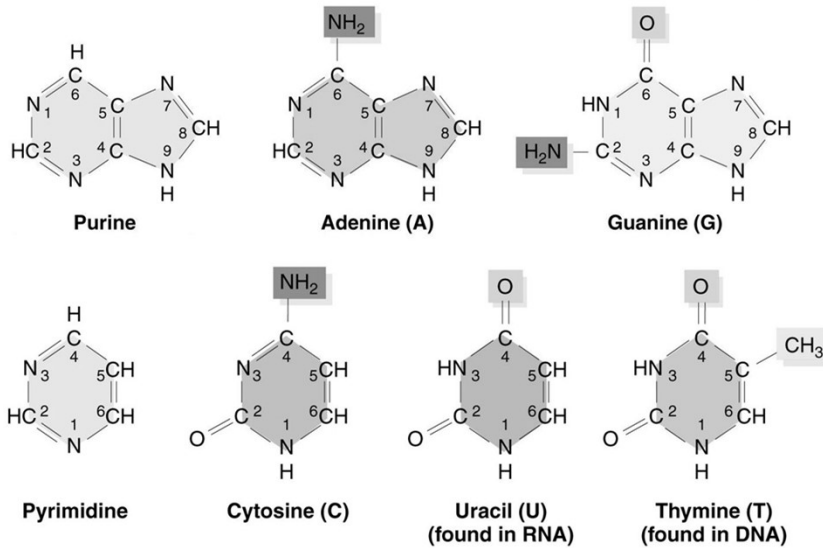
Energy molecules

ATP – Adenosine tri-phosphate

GTP – Guanosine tri-phosphate

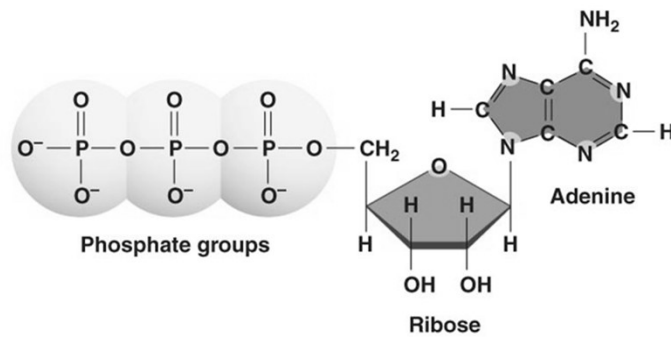
1 ATP	– 1 ATP
1 GTP	– 1 ATP
1 NAD.2H	– 3 ATP
1 FAD.2H	– 2 ATP

Nitrogen Bases



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ATP Molecule



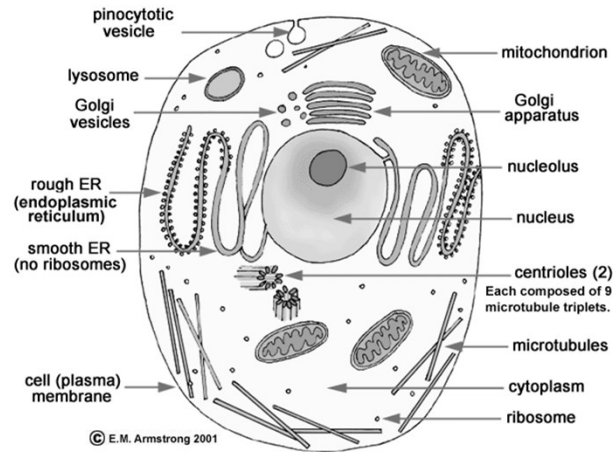
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Glycolysis

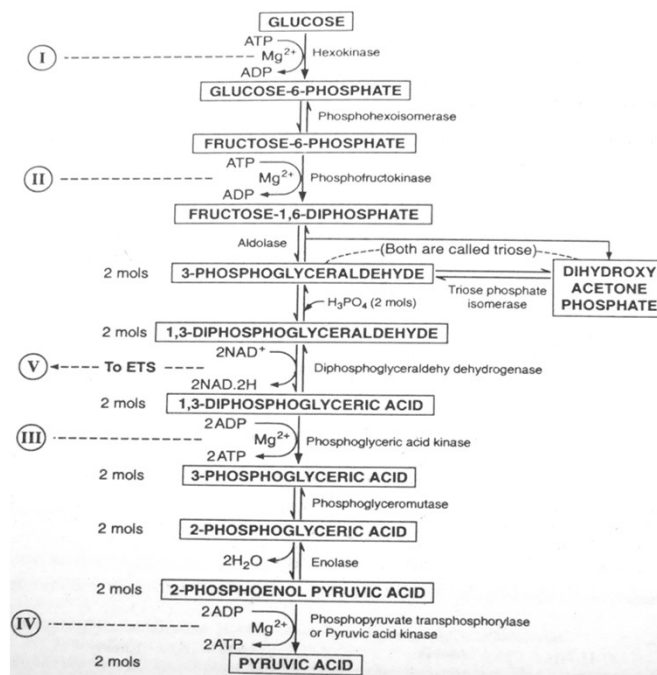
- It is the sequence of 10 enzyme-catalyzed reactions that convert 1 molecule of Glucose in 2 molecules of Pyruvic Acid.

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Structure of an animal Cell

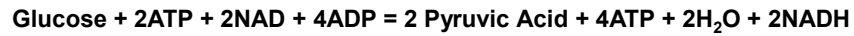


Glycolysis / EMP (Embden Meyerhof & Parnas) Pathway

I Phase
Preparatory PhaseII Phase
Payoff Phase

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Stoichiometry of Glycolysis

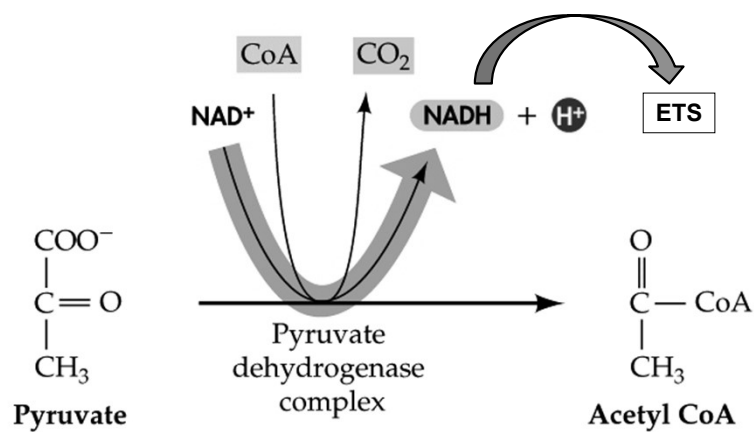


1. Glucose is oxidized to Pyruvic Acid
2. NAD is reduced to NADH
3. ADP is phosphorylated to form ATP

Net Gain of 2ATP Molecules

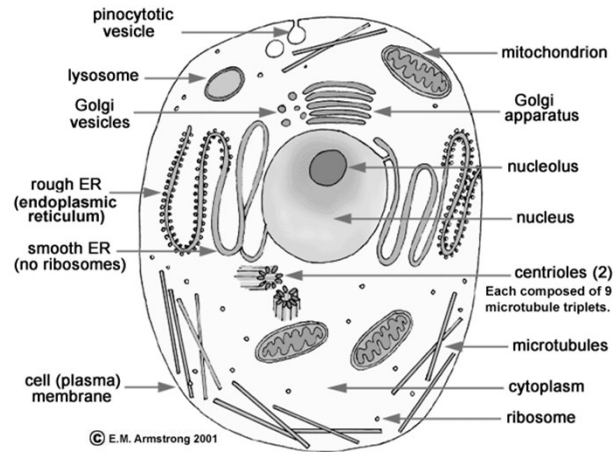
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Pyruvate Oxidation

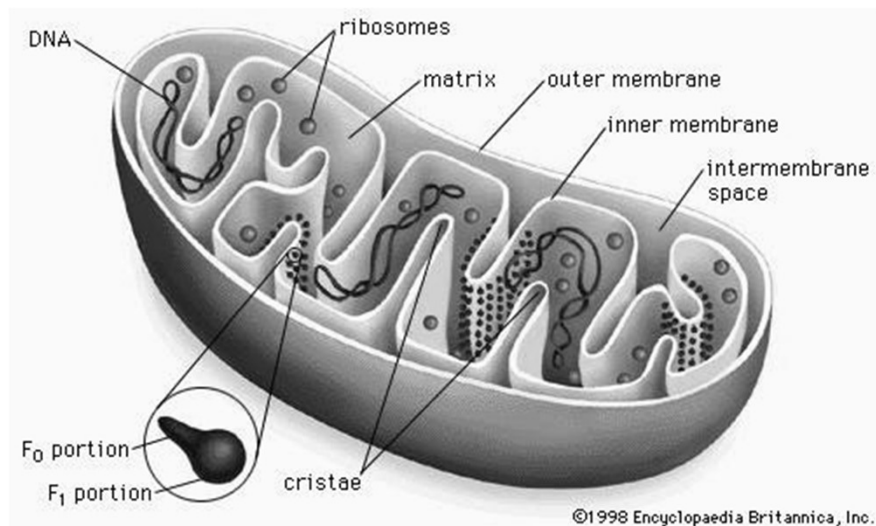


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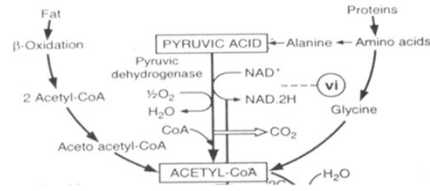
Structure of an animal Cell



Mitochondria

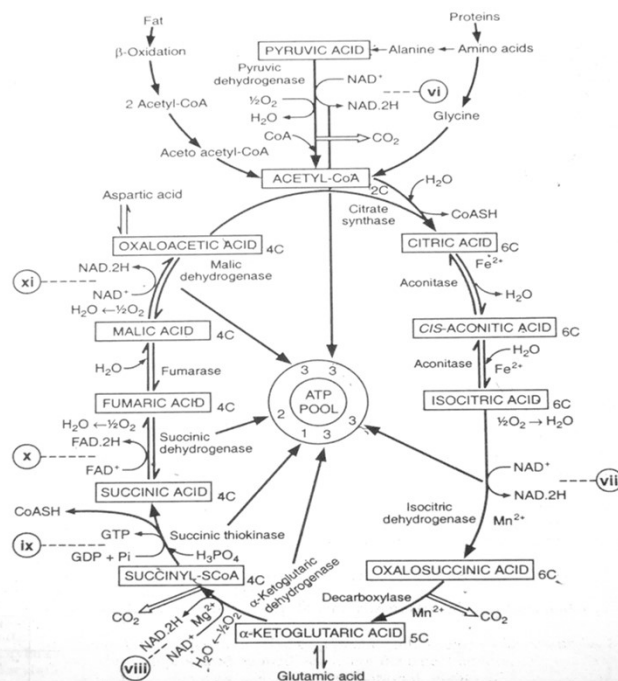


Kreb's Cycle / TCA / Citric Acid Cycle



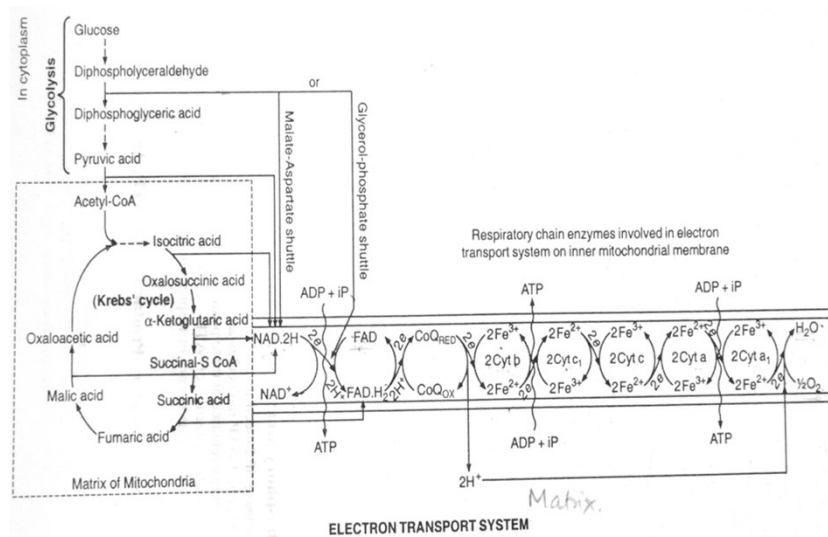
MITOCHONDRIA

Kreb's Cycle / TCA / Citric Acid Cycle



MITOCHONDRIA

Electron Transport Chain

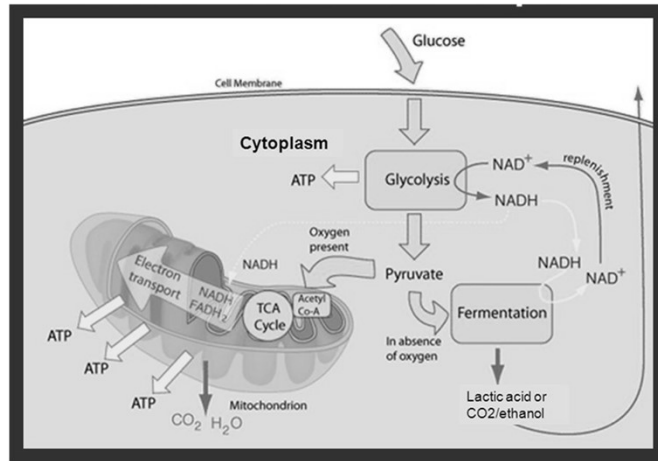


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ATP Production from 1 Molecule of Glucose

1. Glycolysis -----2 ATP-----2 ATP
 2. Glycolysis-----2 NADH₂---(3x2NADH₂)-----6 ATP
 3. Intermediate step-----2 NADH₂---(3x2NADH₂)-----6 ATP
(PA to ACo-A)
 4. Kreb's Cycle 3NADH₂---(3x3NADH₂)---9 ATP
1FADH₂---(2x1FADH₂)---2 ATP
1GTP----- (1x1GTP)-----1 ATP
12 ATPx2Cy-----24 ATP
- TOTAL = 38 ATP**

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Summary

THANK YOU